Explainable Artificial Intelligence for Pulmonary Disease Detection from Chest Radiographs

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***Abstract*—The integration of Explainable Artificial Intelligence (XAI) with deep learning has revolutionized medical image analysis, which is now able to yield accurate and interpretable disease diagnosis. In spite of significant advances in the interpretation of chest X-ray (CXR) for diseases such as tuberculosis, pneumonia, and COVID-19, a number of issues remain to be solved in the fields of explainability, dataset diversity, and clinical trust. This review has comprehensively reviewed 27 research papers (2020– 2025) that focus on the use of XAI in pulmonary imaging. It has mapped out the methods, datasets, visualization tools, interpretability results, and performance metrics of the papers. Moreover, it points out the gaps in research concerning the identification of diseases in multi-class, generalizability of models, and validation of the clinical environment, thus generating the next-stage ideas to move towards hybrid CNN–Transformer architectures and clinician-centered explainable AI.**

Keywords—Explainable AI, Deep Learning, Vision Transformers, Chest X-ray, Pulmonary Disease, Medical Imaging, Grad-CAM, Clinical Trust.

I. INTRODUCTION

One of the major contributions of Artificial Intelligence (AI) to the healthcare domain is the improvement of medical image analysis. This improvement can be noticed from the detection of pulmonary diseases through radiographs of the chest (CXRs) and CT scans that has been significantly enhanced by AI [1]. Various deep learning architectures such as Convolutional Neural Networks (CNNs), and their derivatives like ResNet, DenseNet, and EfficientNet, have been proven to achieve high accuracy in the identification of the lungs diseases including tuberculosis (TB), pneumonia, and COVID-19 [2], [3]. Nevertheless, those robust yet complex models have never been clinically used due to their internal mysteriousness. It is a known fact that these ”black boxes” do not provide any explanation or rationale for their output other than the result itself. Such an absence of transparency becomes the biggest obstacle in a field with serious consequences like medicine, where it is understood that knowing the reason for ’why’ is just as important as the diagnosis itself [4]. Having no possibility to challenge the inner workings of the model may result in the clinicians’ distrust of the technology, in case they are the ultimate decision-makers for patient care [5–7].

Explainable AI (XAI) is a field that has been recognized as a necessity in order to confront the ”black-box” issue raised by AI models. The primary objective of XAI is to formulate such techniques that acquire transparency and interpretability in models, which means that doctors can understand, trust, and efficiently control AI-powered diagnostic instruments. XAI, by giving a picture of the aspects and the logic that resulted in a particular prediction, can be the link between the performance of the algorithm and its utility in practice. In fact, this not only builds up confidence but also allows for error analysis, facilitates the identification of model biases, and ensures that the model picks medically relevant features rather than coincidental ones in the data to make its decisions [8]. The use of XAI has now been extended to area beyond mere classification, such as complex issues like risk prediction for lung cancer [9].

Firstly, the use of XAI methods in pulmonary imaging has undergone a quick transformation. Initially, the research was largely focused on the utilization of post-hoc visualization techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) and Layer-wise Relevance Propagation (LRP) in standard CNN architectures [10]. Such methods produce heatmaps which indicate the areas of a CXR that had the most significant impact on the model’s prediction. Although these post-hoc explanations have their merits, they are generated after the model has taken its decision and, therefore, may not always represent the model’s internal logic accurately.

Inherently interpretable models and hybrid architectures have been the focus of the recent shift in the field. The introduction of Vision Transformers (ViTs) is one of the major changes that the field had. Due to their integrated self-attention mechanisms, ViTs can quite straightforwardly give the attention maps that show which parts of the input image the model looked at most, thus supplying an even more intrinsic form of explainability [11–13]. Besides, a hybrid model, which features the local feature extraction abilities of CNNs and the global context understanding of ViTs, is getting more and more successful in both improving the system’s accuracy and making it interpretable [14].

Despite these advancements, there are still plenty of problems. The majority of the studies have limited their focus to binary or limited multi-class classification, thus, they

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haven't dealt with the problem of differential diagnosis complexity in the case of a real-world clinical setting where multiple pathologies can also be present. The problems of the model's generalization over different patient demographics and imaging equipment are still there because mostly small and homogeneous datasets have been used [15], [16]. Most importantly, a great necessity for more strict, human-centered evaluation of XAI systems is at the forefront. The explanation cannot be only technically correct; it must also be efficient and comprehensible for the clinician and thus it should lead to the improvement of diagnostic accuracy and the decisionmaking confidence [7]. The present article offers a detailed and comprehensive survey and analysis of the literature in the area of XAI in pulmonary disease detection from 2020 to 2025. It maps the research evolution, identifies important research gaps, and points out the future directions of clinically robust and trustworthy AI development.

II. METHODOLOGY OF LITERATURE SELECTION

In order to carry out a systematic and comprehensive review, we extended our search for publications to various academic databases of which are IEEE Xplore, SpringerLink, ScienceDirect, MDPI, PLOS ONE, and the arXiv preprint server. The search was meant to grab the first-rate research that talks about the combination of XAI and pulmonary imaging. Our search strategy utilized the mixture of the keywords and the corresponding variants like "Explainable AI," "XAI," "interpretable deep learning," "chest X-ray," "CXR," "pulmonary disease," "lung disease," "tuberculosis," "pneumonia," and "COVID-19."

The process of selecting works was the result of the use of a predetermined set of inclusion and exclusion features which helped to ensure the relevance of the chosen papers and their quality to be included in this survey. Inclusion Criteria:

- Research papers should have been officially published or made available as preprints within the period of January 2020 to December 2025.
- The main point of the research should be the use of XAI techniques for the identification or classification of pulmonary diseases.
- Studies should be based on the use of chest radiographs (CXRs) or Computed Tomography (CT) scans as the main source of the images.
- The article should offer a detailed explanation of the model framework, the XAI method used, and the visual or quantitative output of the interpretations.

Exclusion Criteria:

- Research works that applied deep learning to detect lung diseases but lacked any explanation or interpretation of the results.
- Works that dealt only with the theoretical side of XAI without referring to a particular application in medical imaging.
- Review articles or surveys that did not report new empirical results and were only considered if they offered a distinctive and comprehensive synthesis relevant to our scope.

The screening process was conducted in three stages. Initially, we looked at the publication titles and abstracts to remove those studies which were obviously irrelevant. Then, we decided the eligibility of the papers by reading the full text and checking them against the inclusion and exclusion criteria. We also got a bunch of information from each paper such as: the model architecture, dataset(s) used, specific XAI technique(s), the target disease(s), the main contributions, and the findings. Lastly, we combined the extracted information to determine the main themes, trends, and research gaps. Our systematic approach led us to the 27 pivotal papers that constitute the foundation of this survey, thus providing a balanced and comprehensive overview of the evolution and current state of XAI in pulmonary disease detection.

III. EVOLUTION OF EXPLAINABLE AI IN PULMONARY IMAGING(2020-2025)

The use of XAI in pulmonary imaging has been majorly, and quite fast, through three stages over the last five years. The first stage is the early phase concentrating on basic posthoc techniques. The second stage is a transition phase where hybrid models are getting prominent, and the third stage is the maturity phase where there is a significant clinical integration and the dominance of inherently interpretable models.

A. Early Phase (2020-2021): Foundational CNNs with Post-Hoc Explanations

The period presented here marks the very first wave of XAI research in which post-hoc visualization techniques were applied to CNN architectures that were already well-established. The chief aim was to make these high-performing but blackbox models transparent by using retrofitting techniques. GradCAM [10] and its variants, like Grad-CAM++ [17], became the procedures most commonly used. These techniques generate class-discriminative localization maps or heatmaps, which show the pixels in the input image that had the most impact on a certain classification. From the point of view of doctors, thus, it represented a necessary, though very simple, first attempt of a model's decision grasping. They could check with their own eyes whether the model was zeroing in on the correct and relevant regions of the lung, for example, areas of consolidation for pneumonia or opacities for COVID-19.

B. Transition Phase (2022-2023): Hybrid Architecture and Multi-Method Explanations

The transition phase of the research community to the limitations of the early phase was clearly marked by a shift to more advanced model architectures such as multi-scale CNNs [18] and the employment of a wider range of XAI techniques. One of the major advancements was the creation of hybrid models that leveraged the benefits of both CNNs and Transformers. For instance, Okolo's IEViT [13] fused a CNN for local feature extraction with a ViT for capturing global, long-range dependencies in the image. By this means, the researchers intended to furnish the model with a more comprehensive understanding of the CXR, thereby bringing about both better classification and higher quality explanations, the latter being the ViT's attention maps. The use of ensemble models with fine-tuned hyperparameters also became more popular [19], [20].

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C. *Maturity Phase (2024-2025): Self-Explainable Models and Clinician-Centered Evaluation*

The last stage is mainly defined by these two major features: the transition to inherently interpretable or ”self-explainable” models, and a much stronger focus on evaluating XAI systems in a clinical context. In this phase, Vision Transformers have been the main architecture. Their self-attention mechanism is no longer considered just a part of a hybrid model but as a powerful, single, engine for both classification and explanation. The studies by Singh et al. [2] and Vanitha et al. [11] have shown the high accuracy of ViTs for pneumonia and TB detection, respectively. At the same time, the attention maps were used as the main source of interpretability.

Importantly, the evaluation focus has changed from algorithmic metrics only to human-centered ones. Researchers have come to the understanding that the ”best” explanation is the one that helps a clinician the most. Prinster et al. [7] carried out an essential study determining the impact of different types of AI explanations on physicians' diagnostic performance and their trust in the AI.

IV. Evaluation Metrics for XAI in a Clinical Context

Assessing XAI systems in medical imaging is a sophisticated, multi-layered issue that cannot be measured simply by standard performance metrics such as accuracy or AUC. An explanation that is clinically helpful should be not only technically correct but also consistent, clear, and a trustworthy source for a human expert. The evaluation criteria can be roughly divided into quantitative, trustworthiness-based measures and qualitative, human-centered evaluations.

Quantitative and Faithfulness Metrics: These measurements are intended to be an unbiased instrument to quantitatively measure the explanation quality and the reliability of the XAI-derived explanations.

- Visual Alignment and Pointing Game: One of the standard ways is to calculate the spatial overlap between the heatmap generated by the XAI and a ground-truth segmentation mask of the pathology annotated by a radiologist. Overlap metrics such as Intersection over Union (IoU) or the Dice coefficient are typically used. A higher overlap indicates that the model is paying attention to the correct pathological region, thus localizing the disease indicators visibly [22]. Besides, more advanced methods qualify for effective localization through weakly supervised approaches [23].
- Faithfulness and Robustness: Faithfulness metrics put forward the idea that the explanation is the model's decision-making process which is the main source of the information. The main method of perturbation analysis can be used in a situation when the XAI identifies the most ”important” regions, and then these regions are masked or altered in the input image. By removing these important regions, if the explanation is faithful, the model's prediction confidence should decrease markedly or the predicted class should change [4].

Human-Centered and Qualitative Metrics: In the end, the worth of an XAI system is mainly judged by the effect it has on the clinical user. These kinds of evaluations usually imply studies with radiologists and clinicians aimed at figuring out the practical explanatory utility.

- Clinician Trust and Reliance: Maybe it is the most important measurement of all the ones considered. Researchers question or watch clinicians so as to get their opinion about the level of trust in AI predictions when explanations are given. Prinster et al. [7] presented comprehensive research results demonstrating that different explanation types (for instance, heatmaps vs. textual explanations) influence physician trust to varying degrees.
- Diagnostic Performance and Efficiency: The main goal is to make patient outcomes better. Hence, first, the evaluations check whether the XAI tools lead to better diagnostics by clinicians and fewer errors (false positives/negatives) and, second, the time for interpreting an image (decision latency) is shortened.

The thorough evaluation of an XAI system should be based on the blend of quantitative and qualitative metrics presented here. Metrics of faithfulness are very important for developers as it helps them to maintain technical robustness but human-centered evaluations are necessary to authenticate real-world clinical utility and safety of the system.

V. Research Gaps And Persistant Challenges

Although the advances have been very fast and have been recorded by this review, there are still major research gaps and challenges that hinder the widespread and stable use of XAI systems in clinics for pulmonary disease detection. Solving these problems is essential for the area to transfer from university research to practical use in hospitals.

- Limited Generalization and Dataset Bias: A longstanding problem has been the dependency on small, uniform, and frequently publicly available datasets [15]. The models that are trained on these datasets might yield good results during testing; however, they do not have the capability to generalize to different patient populations and the imaging protocols that have variations, which are typical for real clinical environments.
- Predominance of Single-Disease Classification: Most of the research works have been concentrating on classification of binary or limited multi-class tasks [11], [21]. Nevertheless, it is required that models classify several diseases at the same time by means of methods such as fine-tuned transfer learning [24]
- The Fidelity of Post-hoc Interpretability: Post-hoc methods such as Grad-CAM have largely been helpful, however, their explanations represent a rough understanding of the model's reasoning and, therefore, might not be very consistent with the internal decision-making process. This has been the main reason for the move towards inherently interpretable or self-explainable models [25].

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TABLE I. SUMMARY OF KEY STUDIES IN XAI FOR PULMONARY IMAGING (2020-2025)

Study	*Phase*	*Architecture/ Model*	*Key Contribution*
Panwar et al. [10]	Early	VGG-19 + Grad-CAM	Demonstrated heatmap visualization for rapid COVID-19 detection.
Ali & Khan [12]	Early	Vision Transformer (ViT-COS)	Pioneered ViT for CXR analysis with intrinsic attention-based explainability.
Bhandari et al. [21]	Transition	Lightweight CNN	Applied a suite of XAI tools (Grad-CAM, SHAP, LIME) for multi-disease classification.
Okolo [13]	Transition	IEViT (CNN+ViT Hybrid)	Developed a hybrid model combining local (CNN) and global (ViT) features.
Singh et al. [2]	Maturity	Vision Transformer (ViT)	Pneumonia detection with ViTs demonstrated a high level of efficiency.
Prinster et al. [7]	Maturity	Physician Trust Study	Analyzed with great depth how different XAI explanations influenced the trust of physicians.
Anderson et al. [14]	Maturity	Hybrid CNN+ViT	Proved that physicians' diagnostic accuracy is enhanced when they are assisted by XAI deep learning.

- Computational Cost and Workflow Integration: Advanced deep learning models, especially large Vision Transformers and complicated ensemble models, tend to be computationally intensive. The additional overhead of producing explanations with methods such as GradCAM, SHAP, or LIME, can, therefore, be a substantial contributor to inference time [16], [20].

VI. DISCUSSION

The inclusion of XAI in pulmonary imaging over the years 2020 to 2025 can be seen as a progression of a technology that is quickly becoming developed, gradually departing the initial demonstration stage and dealing with intricate issues of practical use in clinics. The change, as it were, is very significant: it is not only going from basic CNNs with post-hoc Grad-CAM heatmaps to advanced hybrid CNN-Transformer architectures and self-explanatory models, but also it is quite a different thing altogether to a different level of the problem of how to create AI which can be trusted in medicine.

One of the major changes has been the rise of Vision Transformers. Their capacity to acquire global context makes them extremely suitable for chest radiographs interpretation, where subtle, widespread patterns may be signs of a disease. While CNNs are good at finding local features such as textures and edges, ViTs can establish long-range relationships with the whole image, just like a radiologist doing a holistic review. ViTs' attention maps provide a more direct and thus, potentially more convincing, way of understanding the model compared to the methods applied after the fact. The best models seem to be hybrids that use the advantages of both types: CNNs to keep the local structures of the image and ViTs to gather the global context, as evidenced by the study of physician accuracy improvement by Anderson et al. [14].

Another major theme revolves around the explanations themselves that have been progressively refined in detail. Initially, heatmaps showed a very simple "where" the model was focusing. More advanced techniques such as SHAP and LIME [5], [21] are starting to give a "what" by measuring the different image features' contribution. The end objective is to go beyond "why," thus, explanations that are not just visual pointers but are contextual and semantically meaningful. For instance, a perfect XAI system could not only indicate a lung opacity but also explain that it is identifying pneumonia because of the specific characteristics (e.g., location, density, and the presence of pleural effusion) thus, the AI's reasoning is in line with clinical language.

Still, the survey has revealed the most critical point which is that the consensus of the community is that clinical adoption will depend the most on human-centered evaluation. The work of Prinster et al. [7] is a milestone in this respect, showing that the way an explanation is presented can affect a physician's trust and diagnostic behavior to a large extent. It, therefore, brings to light an important issue: the reason for XAI in medicine is not to replace the clinician but to extend the clinician's capabilities. Consequently, the effectiveness of an XAI system should be gauged by the human-AI team's ability to work together rather than only the algorithm tested in isolation.

VII. FUTURE DIRECTIONS

With the advancements and insights gained in the past five years, the coming days of XAI in lung imaging must be a deliberate move to align the remaining disparities between research and clinical practice. The subsequent points indicate the major areas that depict the future of research and development.

- Development of Robust Hybrid Architectures: The future probably depends on hybrid models that effectively combine the advantages of various architectures. Hybrid CNN-Transformer models are particularly promising. CNNs can act as powerful local feature extractors in the early layers, feeding rich, spatially-aware feature maps into subsequent Transformer layers that can model the global context and complex relationships across the entire lung fields. The next step is to design these architectures with explainability as a core design principle, developing methods for integrated XAI that can trace decisions through both the convolutional and attention-based components of the network.

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TABLE II. COMPARATIVE ANALYSIS OF COMMON XAI TECHNIQUES

Technique	*Core Idea*	*Strengths*	*Limitations*
Grad-CAM	Uses class-specific gradient information to localize important regions.	Highly intuitive visual heatmaps; simple to implement for CNNs.	Post-hoc and can be noisy; may not faithfully represent model logic.
LIME	Approximates the model locally with an interpretable surrogate model.	Model-agnostic; provides clear explanations for individual predictions.	Can be computationally slow; explanations may lack stability.
SHAP	Applies game theory (Shapley values) to attribute predictions to features.	Strong theoretical foundation; provides both global and local insights.	Computationally intensive, especially for images; results can be complex to interpret.
Attention Maps	Visualizes the internal self-attention weights from Vision Transformers.	Intrinsic to the model (not post-hoc); captures global, long-range dependencies.	Requires large datasets; attention is not always a direct proxy for explanation.

- Creation of Standardized, Multi-Institutional Datasets: Large-scale, multi-institutional datasets representing clinical diversity from the real world are a necessity for the field to overcome the generalizability problem which has been a major issue of concern. Besides images, these datasets should be rich in structured clinical data and have detailed annotations from several radiologists. A uniform benchmark dataset would make it possible to have more valid comparisons between different models and XAI methods, thus encouraging research that can be reproduced and making progress faster. Such an effort is a move towards this goal as is the case with the initiative proposed by Shah et al. [26] for data-driven approaches..
- Advancing Beyond Heatmaps to Context-Aware Explanations: While heatmaps serve the purpose, they represent a very basic form of a limited explanation. Studies should continue to discover new, more elaborate and context-aware modalities to explain. One possibility might be to use natural language to create reports that describe the findings in medical terms, or counterfactual explanations (”the model would have predicted ’normal’ if this nodule were not present”) or even building interactive systems where clinicians can ask the model to show them its reasoning in more detail. The aim is to establish a conversation between the clinician and AI, instead of just giving the information in one direction.
- Robust Human-Centered Evaluation Frameworks: The findings of Prinster et al. [7] and similar studies should be transformed into clear criteria for humancentered evaluation. The next research should be the conduction of prospective clinical trials that assess the influence of XAI tools on diagnostic accuracy, interreader variability, reading time, and clinician confidence as primary performance indicators. To achieve this, AI researchers and medical professionals need to work hand in hand to sketch out studies that are of clinical significance and have strong statistics.
- Focus on Real-Time, Resource-Efficient Deployment: One of the main goals should be the development of a resource-efficient real-time deployment strategy. To be able to use XAI in clinical practice, it has to be a computational efficient tool. Research on model compression, knowledge distillation, and quantization methods is necessary to develop models that are both lightweight and high-performing and can be deployed on typical hospital IT infrastructures without the need for expensive hardware. In addition, the XAI module should also be optimized for real-time execution to allow easy incorporation into the clinical workflow, which is the idea of the next-generation systems [27].

VIII. CONCLUSION

The study began with an extensive investigation into how Explainable AI has changed in pulmonary disease identification from 2020 to 2025, and it summarized the results of the 27 most influential research works. The major changes have been rapid and have gone a long way. It has gone from using post-hoc visualization on CNNs, to explaining the working of hybrid architectures and Vision Transformers that are inherently interpretable. There has been a lot of improvements in the automation of the diagnosis of pneumonia, tuberculosis, and COVID-19, which are among the most critical diseases, to name a few. Above all, the researchers have gone beyond merely providing the explanations generated by the models and have now started to evaluate the faithfulness, usefulness, and effect of the explanations on the clinical trust.

However, the journey until these models become widely and responsibly used in the clinic is still complicated with issues. The foremost problems of model generalizability, the requirement of multi-disease diagnostic capabilities, and the lack of rigorous, human-centered clinical validation are still the main obstacles to overcome. The ability to close these gaps by inventing robust, efficient, and clinically-aligned XAI systems will determine the future of the field. In fact, the aim goes far beyond just constructing transparent algorithms; it is about the creation of a synergistic partnership between clinicians and AI which, thus, would lead to

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enhanced diagnostic reasoning, better patient outcomes, and the birth of a new era of trustworthy artificial intelligence in medicine.

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